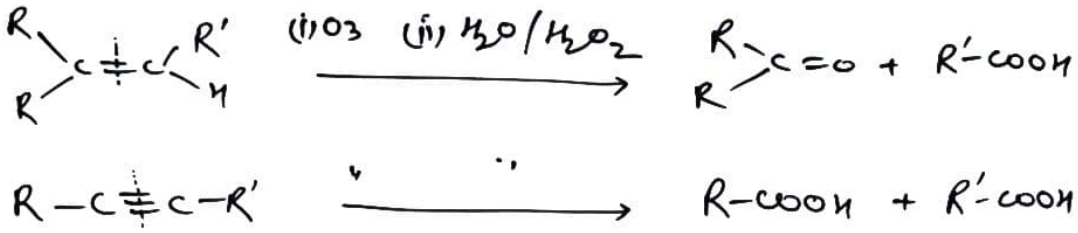
CARBOXYLIC ACID AND THEIR DERIVATIVES

CARBOXYLIC ACID:-

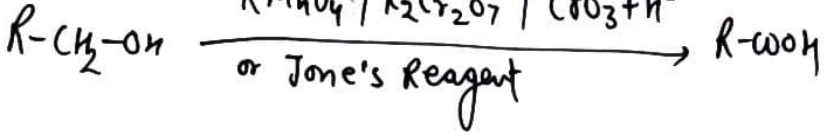
Methods of formation:-

1. By oxidative ozonolysis of alkenes and alkyne

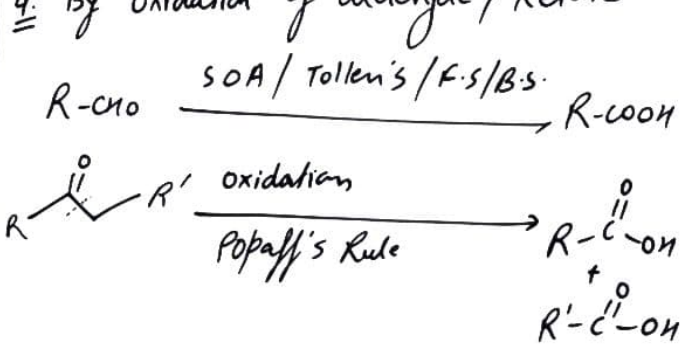


2. By oxidation of alkenes and alkyne with S.O.A.
   → Same as oxidative ozonolysis

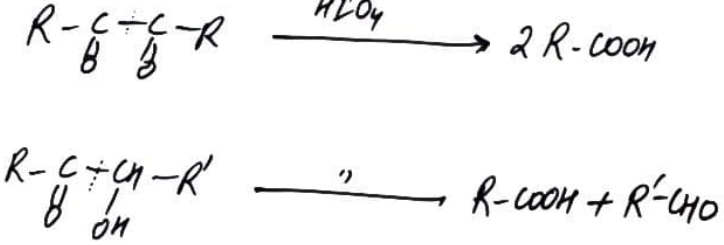
3. By oxidation of 1° alcohols:-



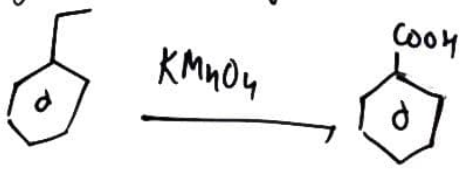
4. By oxidation of aldehyde / Ketone



5. By Oxidative of adjacent dione with HIO_4:-



6. By oxidation of side chain of aro. ring.

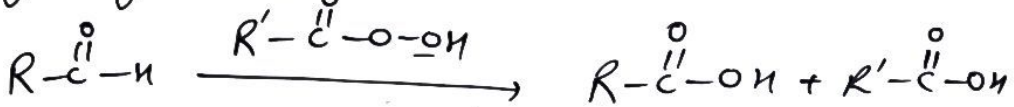


CARBOXYLIC ACID AND THEIR DERIVATIVES

CARBOXYLIC ACID

Methods of formation:-

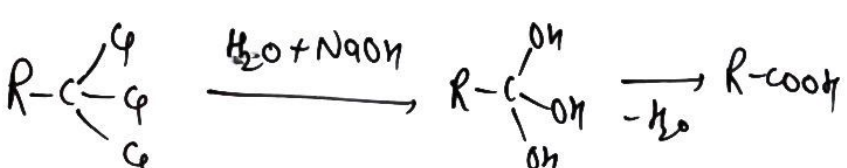
7. By Baeyer-Villiger oxidation of aldehydes:



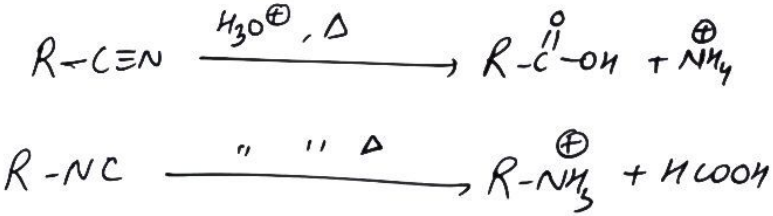
8. By Grignard Reagent:-



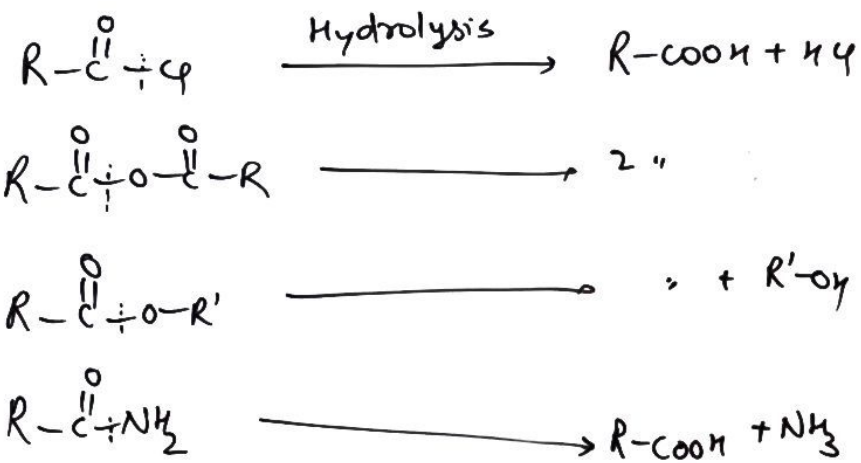
9 By Hydrolysis of 1,1,1-Trihaloalkanes:-



10 By hydrolysis of cyanide & isocyanide:-



(11) By Hydrolysis of acid derivatives:-

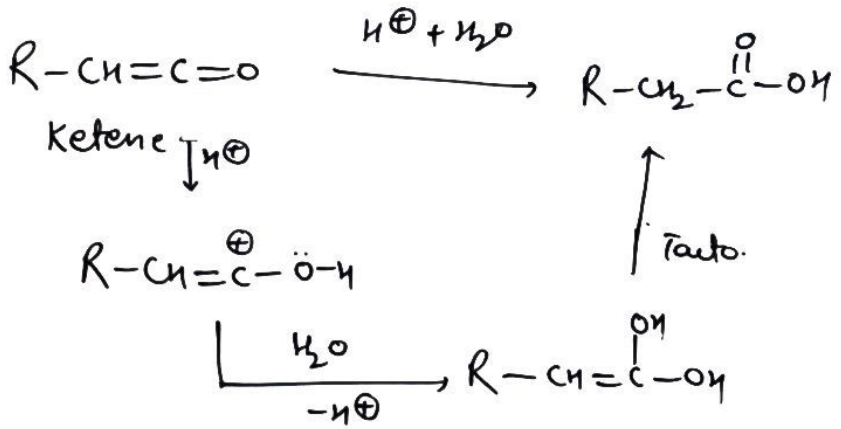


CARBOXYLIC ACID AND THEIR DERIVATIVES

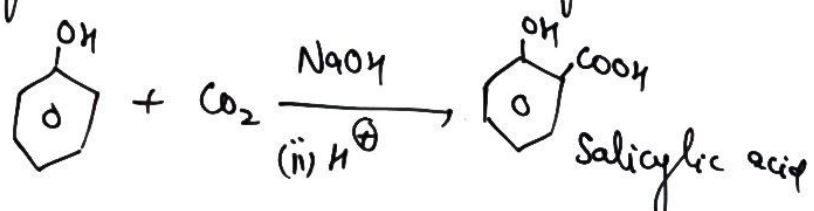
CARBOXYLIC ACID

Methods of formation:-

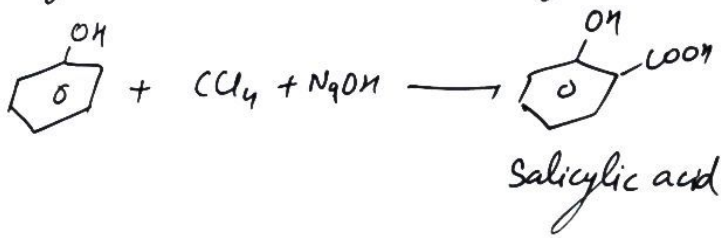
(12) By hydration of Ketene:-



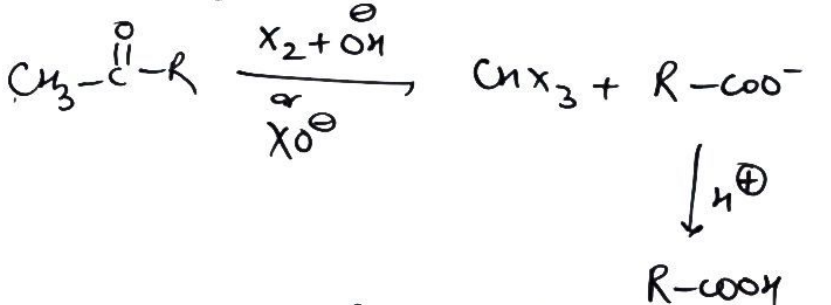
(13) By Kolbe-Schmidt Reaction of Phenol.



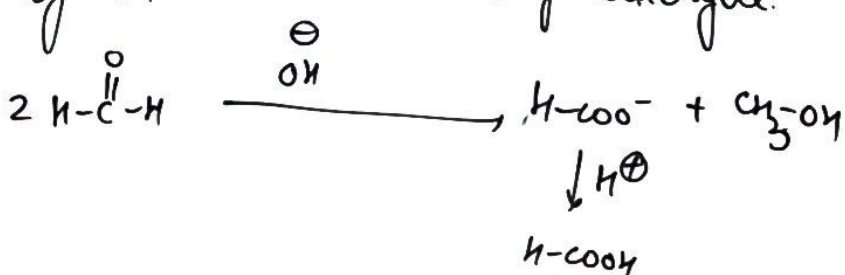
(14) By Reimer-Tiemann Carboxylation.



(15) By Halo form Rx^n



(16) By Cannizzaro Reaction of aldehyde.

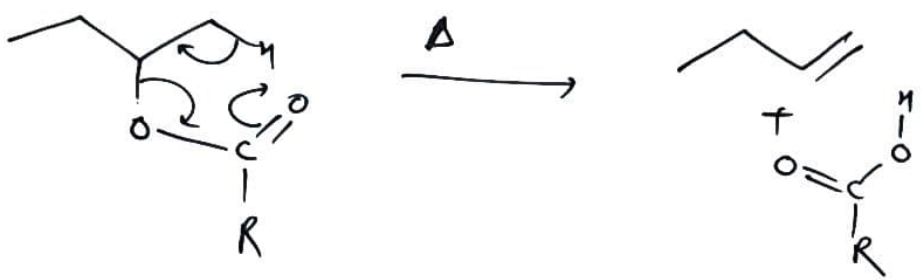


CARBOXYLIC ACID AND THEIR DERIVATIVES

CARBOXYLIC ACID

Methods of formation:-

(17) By Pyrolysis of ester:- (E_i)

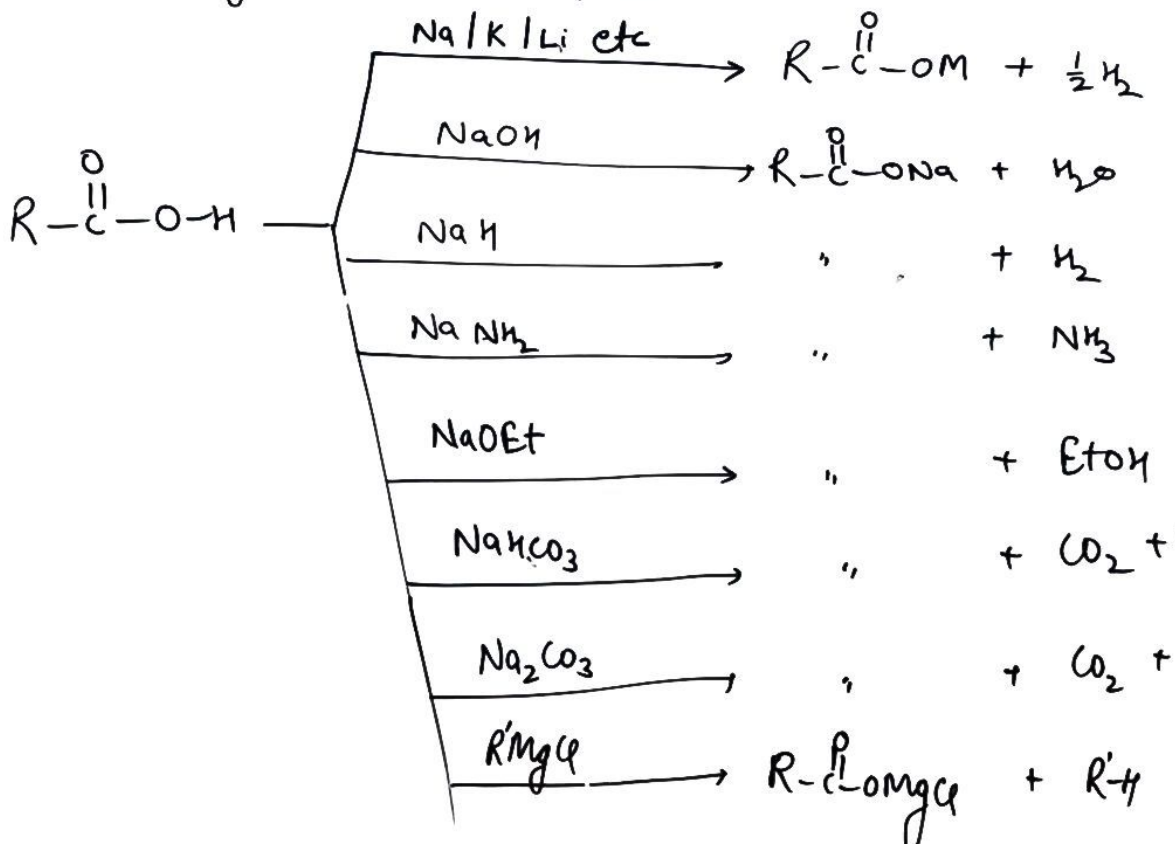


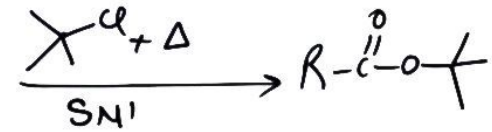
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

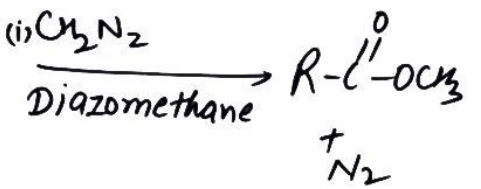
CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

(1) Reaction of O–H bond of –COOH :-





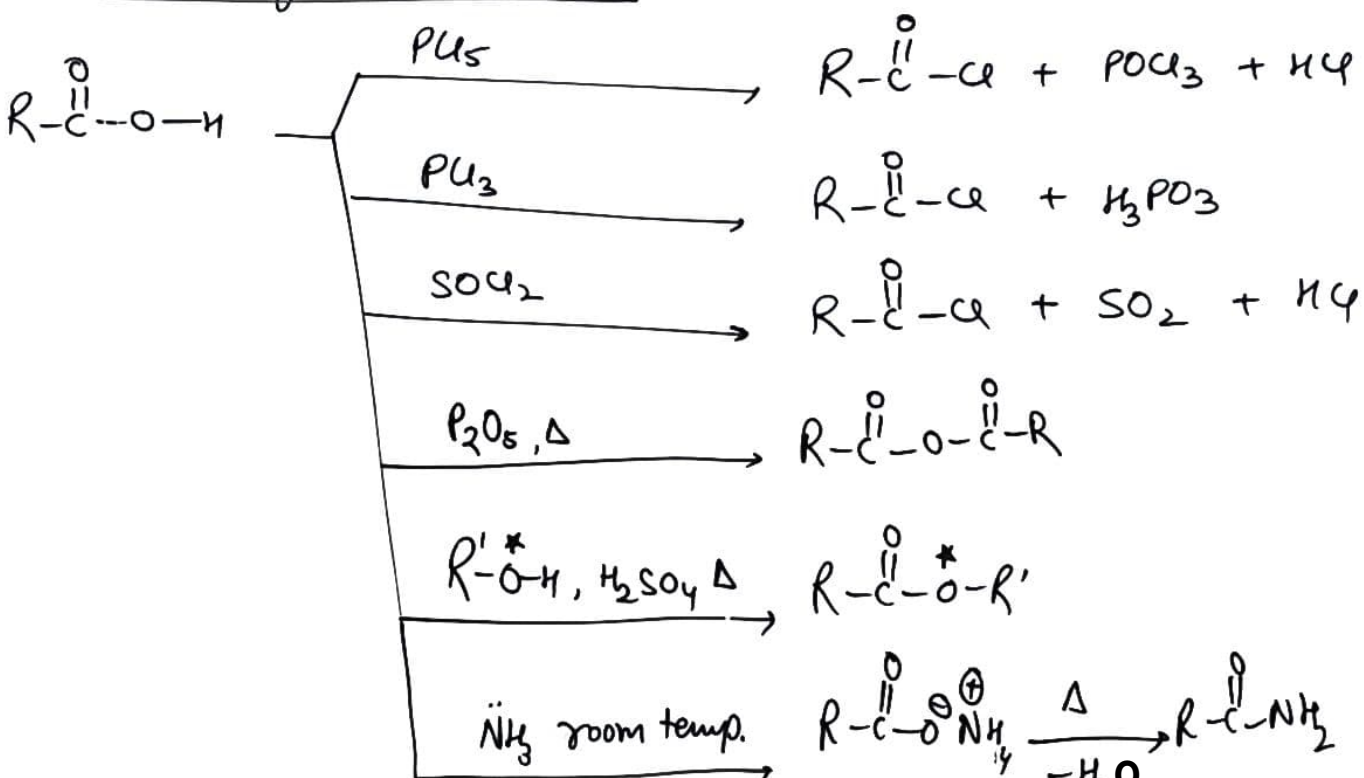


CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

(2) Reaction of c-o bond :-

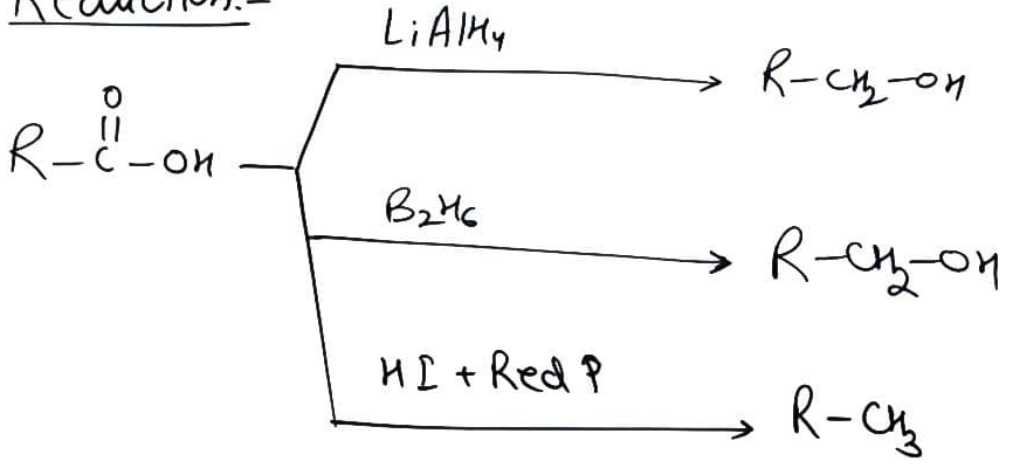


CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

③ Reduction :-



CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

④ Kolbe's electrolysis :- (in ORM-II SFR)

$$\left(R-\overset{O}{\overset{\|}{C}}-OK\right)_{aq} \xrightarrow{\text{electrolysis}} R-R + \underset{\text{anode}}{CO_2\uparrow} + \underset{\text{Cathode}}{H_2} + KOH$$

⑤ Decarboxylation :- (in aromatic comp⇒ Pheno))

$$R-COOH \,/\, R-\overset{O}{\overset{\|}{C}}-ONa \xrightarrow[\text{Soda lime}]{(NaOH + CaO)} R-H + Na_2CO_3$$

$$Ph-COOH \xrightarrow{NaOH + CaO} Ph-H$$

mechanism

$$R-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{\overset{\ominus}{OH}} R-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{O} \xrightarrow{RDS} R^{\ominus} + CO_2$$

$$R^{\ominus} \xrightarrow{H_2O} R-H + \overset{\ominus}{OH}$$

Reactivity of decarboxylation ↑ es on increas stability of carbanion intermediate

$(1^{\circ} > 2^{\circ} > 3^{\circ})$

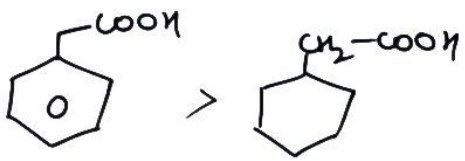
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

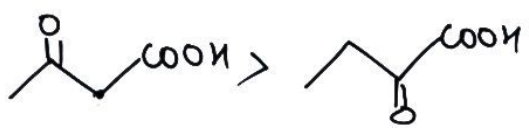
CARBOXYLIC ACID :-

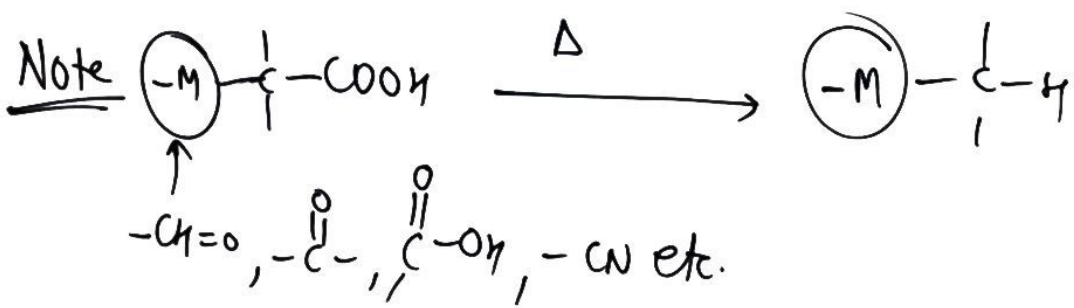
CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

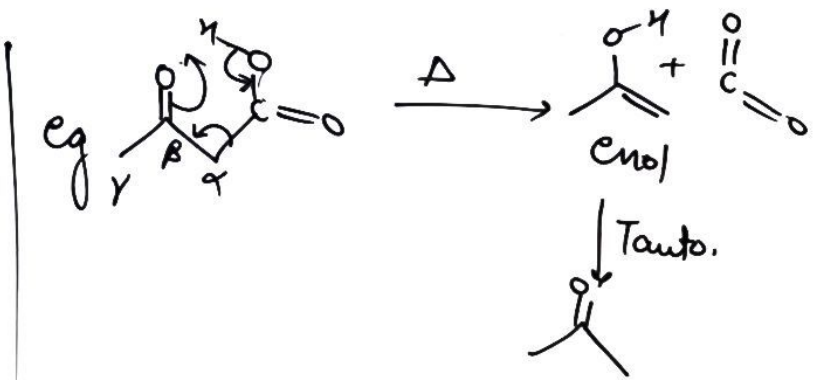
⇒ Reactivity for decarboxylation

(i) $CH_3-COOH > CH_3-CH_2-COOH >$  > 

(ii) 

(iii) 

Note 

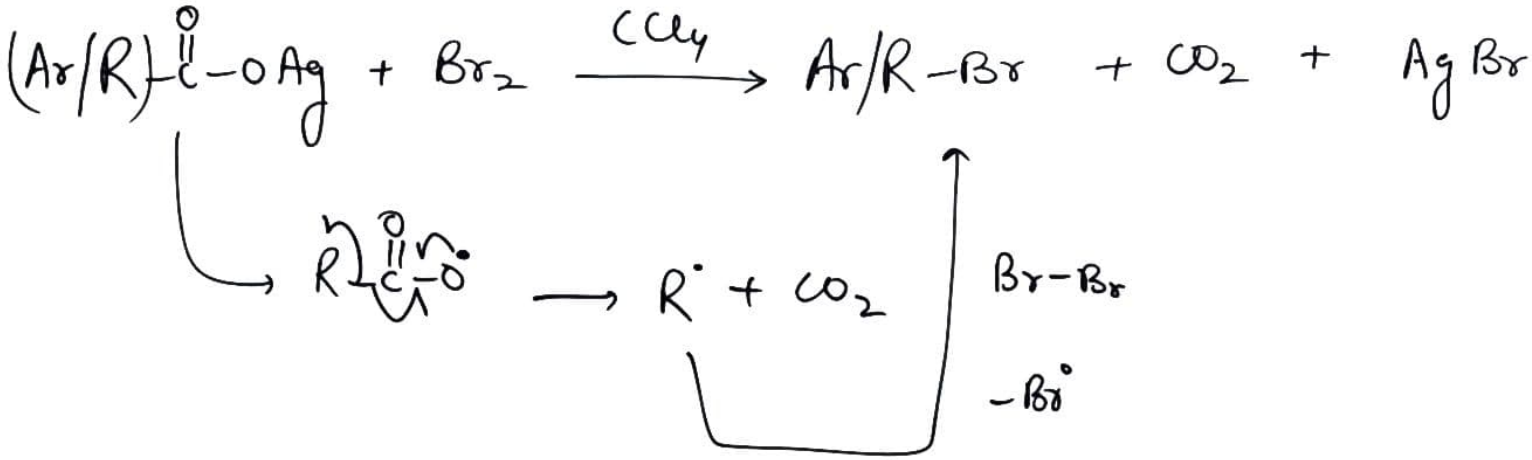


CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

(6) Hunsdeker's Reaction :- (in ORM-II , S_{FR})



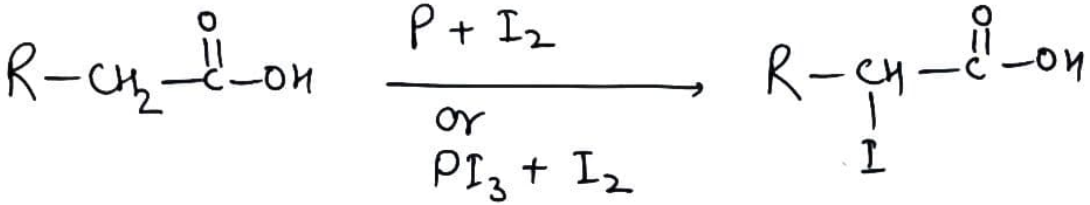
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID:-

CHEMICAL REACTIONS OF CARBOXYLIC ACID:-

(7) Hell- Volhard - Zelinsky (HVZ) Reaction:-

.. It is alpha halogenation reaction **of carboxylic acid**



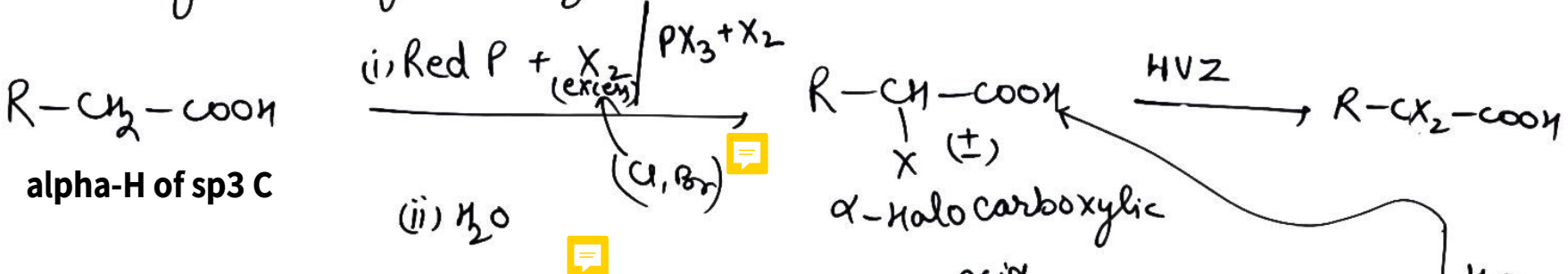
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

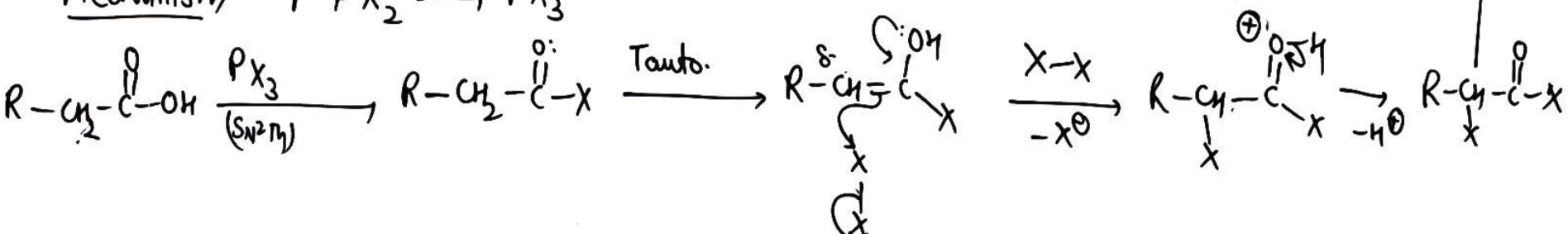
CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

(7) Hell - Volhard - Zelinsky (HVZ) Reaction :-

i- Halogenation of carboxylic acid



Mechanism $P + X_2 \longrightarrow PX_3$

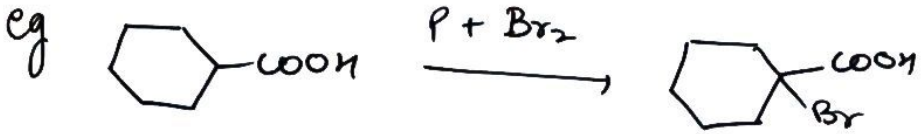


CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

(7) Hell - Volhard - Zelinsky (HVZ) Reaction :-

eg 

eg $Ph-CH_2-COOH \xrightarrow[(HVZ)]{P + Cl_2} Ph-\underset{Cl}{CH}-COOH \xrightarrow{2NH_3\ \Delta} Ph-\underset{NH_2}{CH}-\overset{O}{\overset{\|}{C}}-NH_2$

$\downarrow$ Hydrolysis

$Ph-\underset{NH_2}{CH}-COOH$ α-Amino Carboxylic acid

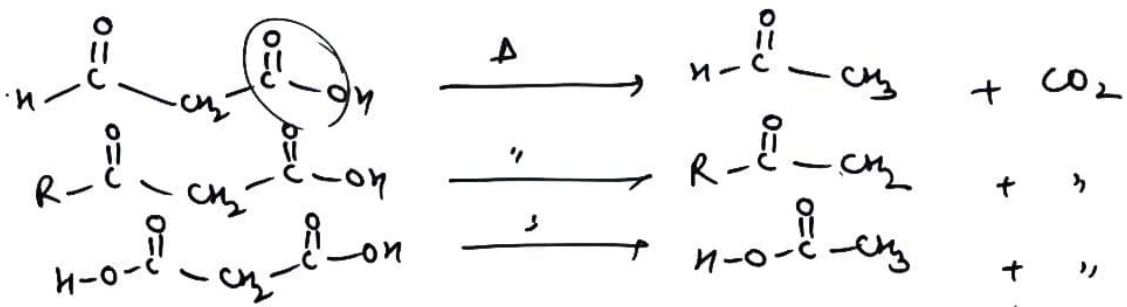
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID:-

CHEMICAL REACTIONS OF CARBOXYLIC ACID:-

(8) Heating effect:-

(a) Decarboxylation of heating:-



CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

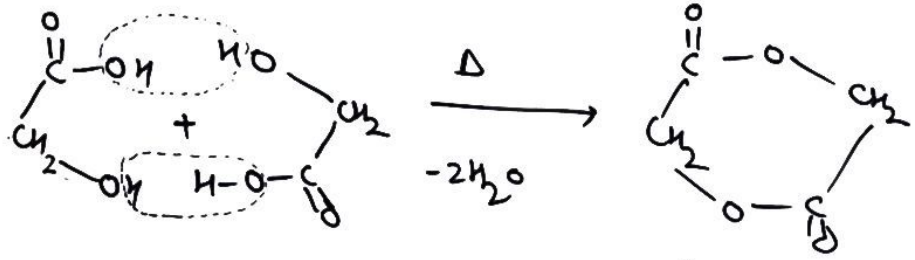
CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

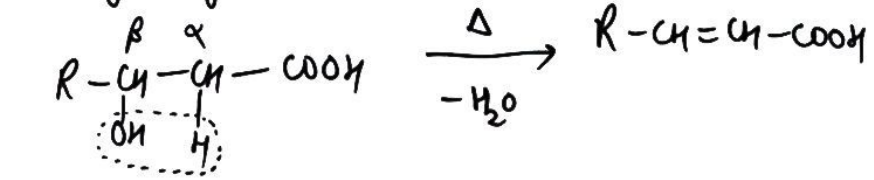
(a) Heating effect :-

(b) Heating of Hydroxy carboxylic acid

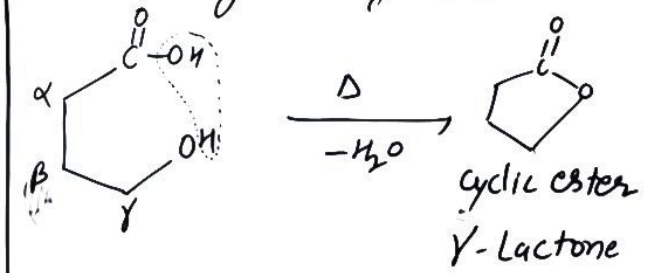
→ α Hydroxy carboxylic



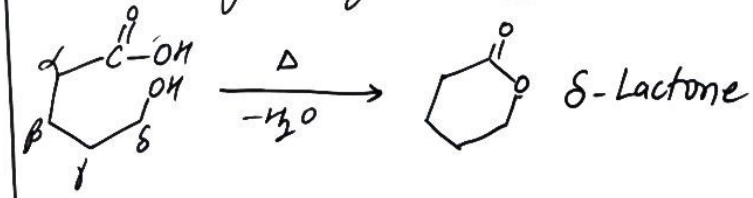
→ β-Hydroxy acid :-



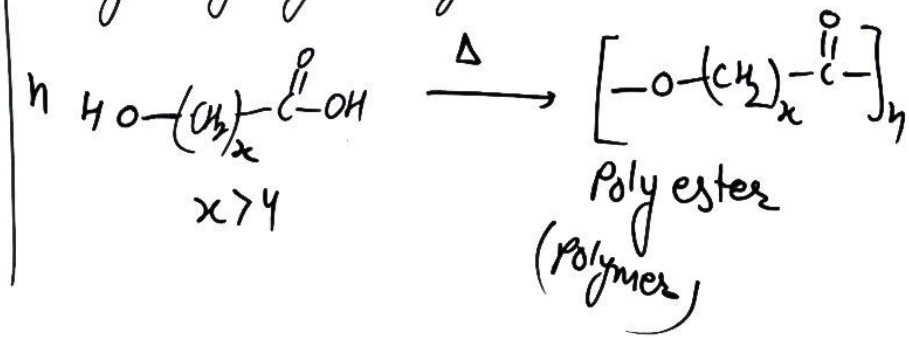
→ γ-Hydroxy carboxylic acid



→ δ-Hydroxy carboxylic acid



→ higher hydroxy carboxylic acid



CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

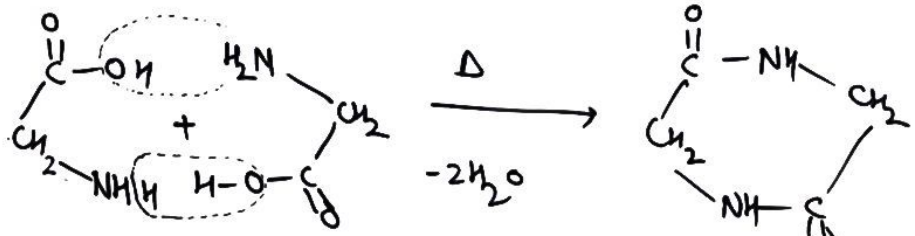
CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

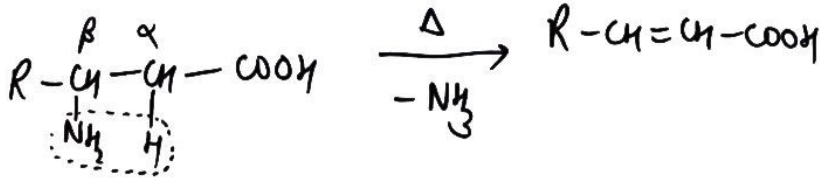
(8) Heating effect :-

(c) Heating of amino carboxylic acid

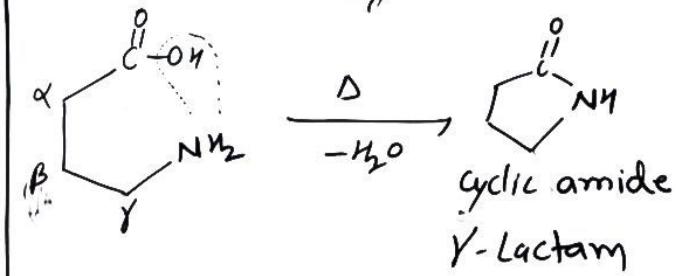
→ α - amino carboxylic



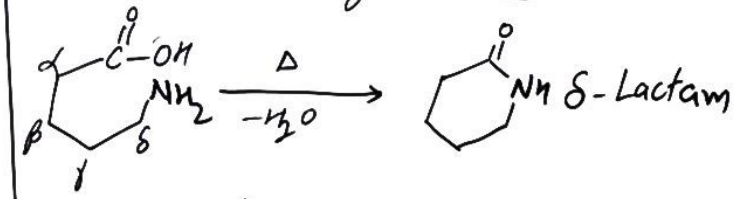
→ β-Amino acid :-



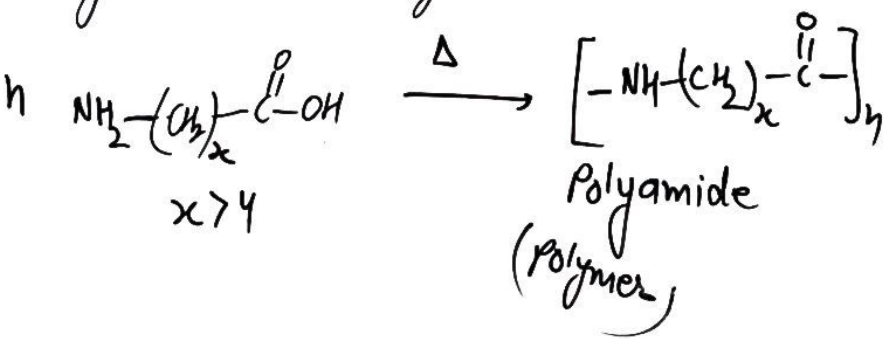
→ γ amino carboxylic acid



→ δ - amino carboxylic acid



→ higher Aminocarboxylic acid



CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

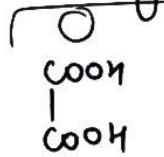
CARBOXYLIC ACID:-

CHEMICAL REACTIONS OF CARBOXYLIC ACID:-

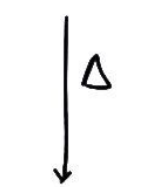
(8) Heating effect:-

(d) Heating of dioic acid:-

O



Oxalic acid

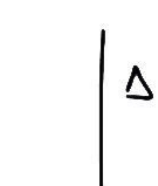


$H\text{-}COOH + CO_2$

M

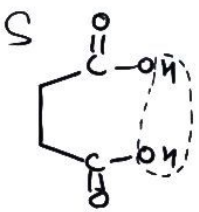


Malonic acid

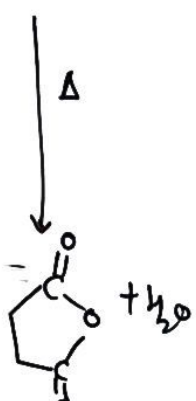


$CH_3\text{-}COOH + CO_2$

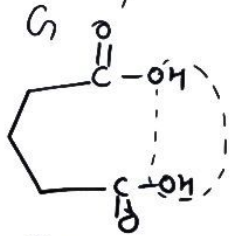
S



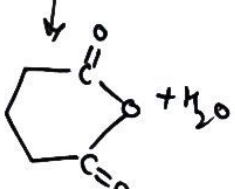
Succinic acid



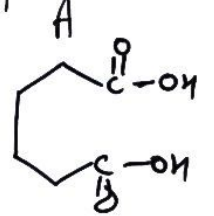
G



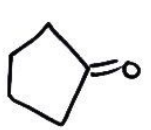
Glutaric acid



A

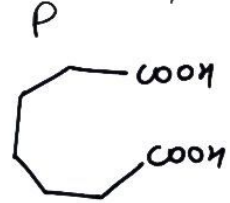


Adipic acid

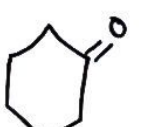


$CO_2 + H_2O$

P



Pimalic acid



$+ CO_2 + H_2O$

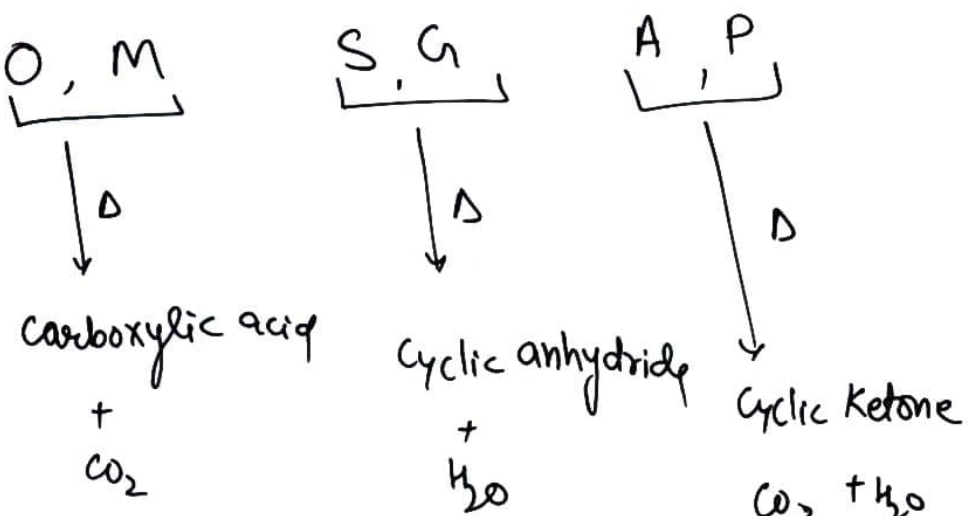
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID:-

CHEMICAL REACTIONS OF CARBOXYLIC ACID:-

(8) Heating effect:-

(d) Heating of dioic acid:-

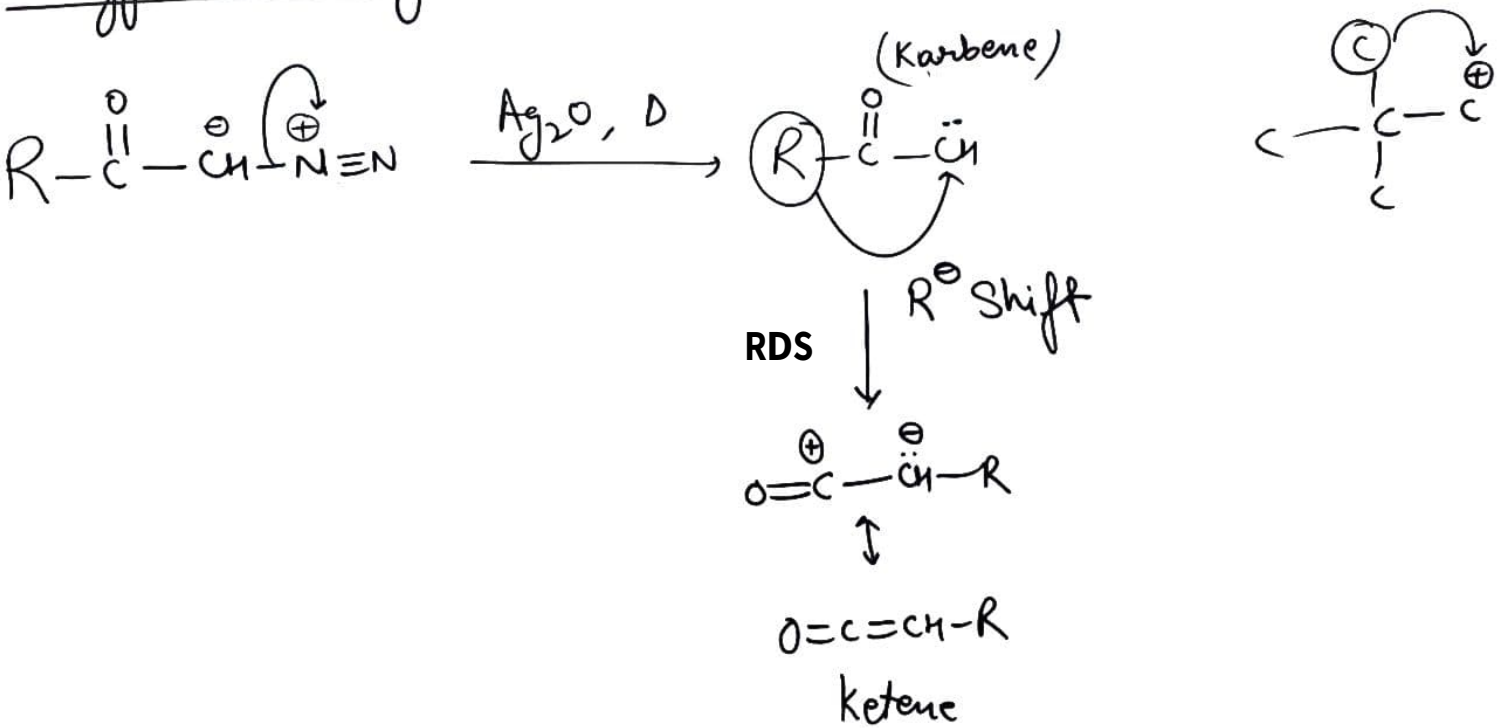


CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

(9) **Wolff - Rearrangement :-**



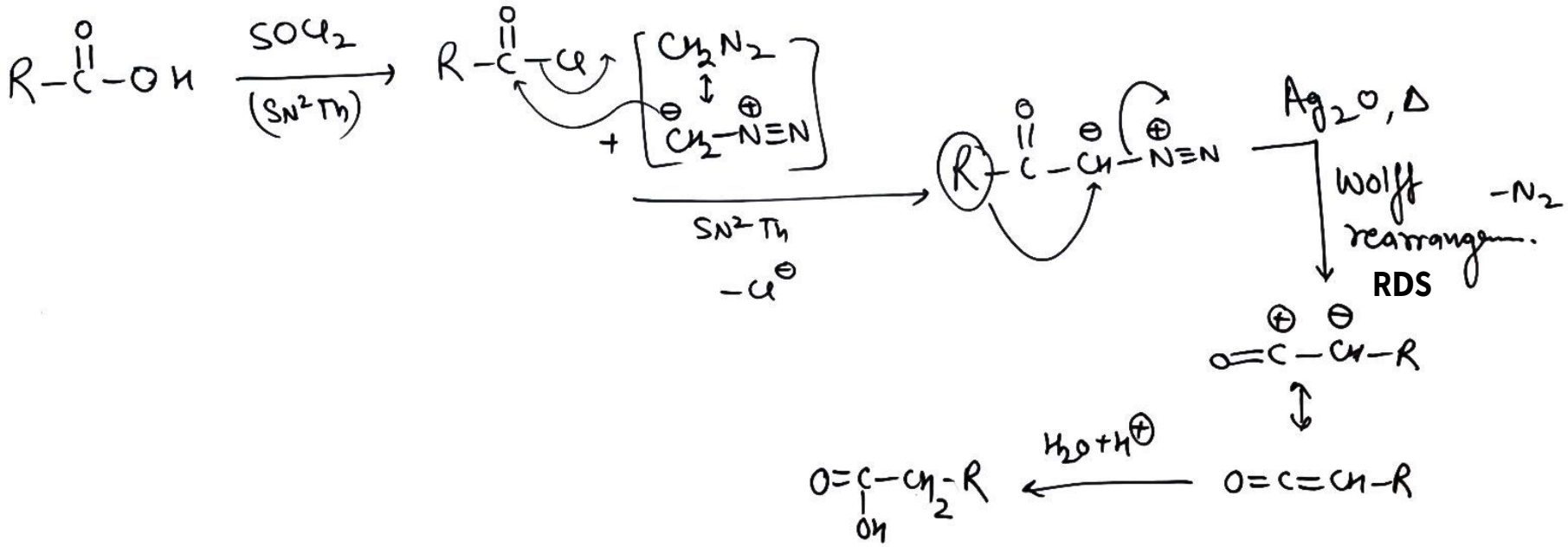
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

(9) **Wolff - Rearrangement :-**

⇒ **Arndt - Eistert synthesis :-** (Reaction for increase one $-CH_2-$)



CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID :-

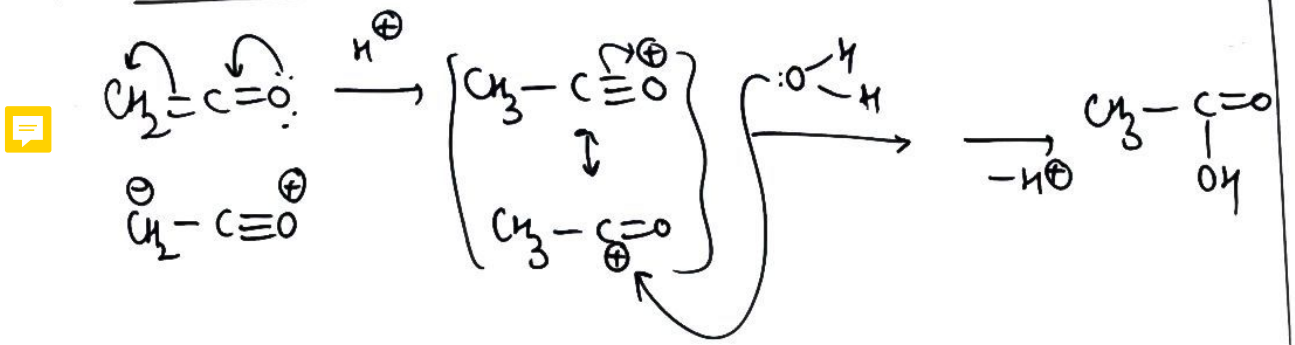
CHEMICAL REACTIONS OF CARBOXYLIC ACID :-

≠ **Reactions of Ketene:-**

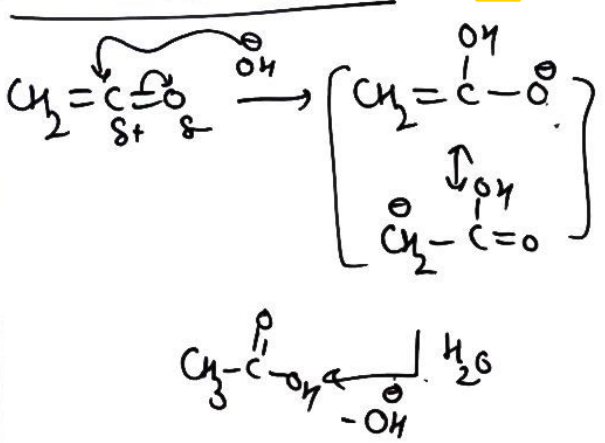
└ can give electrophilic addition as well as nucleophilic addition Rxn

$$CH_2=C=O \xrightarrow[\text{or } H_2O + \overset{\ominus}{O}H]{H_2O + H^{\oplus}} R-\overset{O}{\overset{\|}{C}}-OH \;/\; R-\overset{O}{\overset{\|}{C}}-O^{\ominus}$$

⇒ in acidic medium



⇒ in basic medium:-



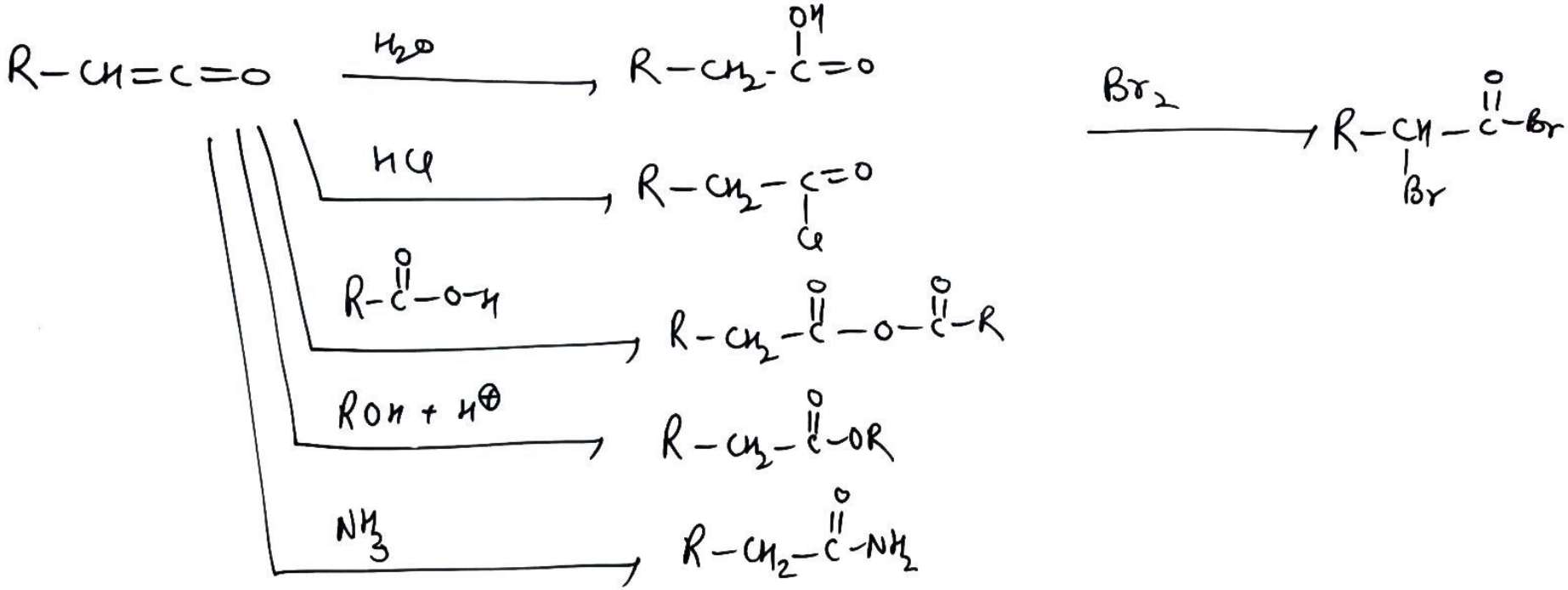
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

CARBOXYLIC ACID:-

CHEMICAL REACTIONS OF CARBOXYLIC ACID:-

≠ **Reactions of Ketene:-**

- can give electrophilic addition as well as nucleophilic addition Rxn



CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

(I) ACID HALIDE

Formation

① $R-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{PCl_5} R-\overset{O}{\overset{\|}{C}}-Cl + POCl_3 + HCl$

$\xrightarrow{PCl_5} R-\overset{O}{\overset{\|}{C}}-Cl + H_3PO_3$

$\xrightarrow{SOCl_2} R-\overset{O}{\overset{\|}{C}}-Cl + SO_2 + HCl$

② $R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R \xrightarrow{PCl_5} 2R-\overset{O}{\overset{\|}{C}}-Cl + POCl_3$

③ $R-\overset{O}{\overset{\|}{C}}-O-R' \xrightarrow{PCl_5} R-\overset{O}{\overset{\|}{C}}-Cl + R'-Cl + POCl_3$

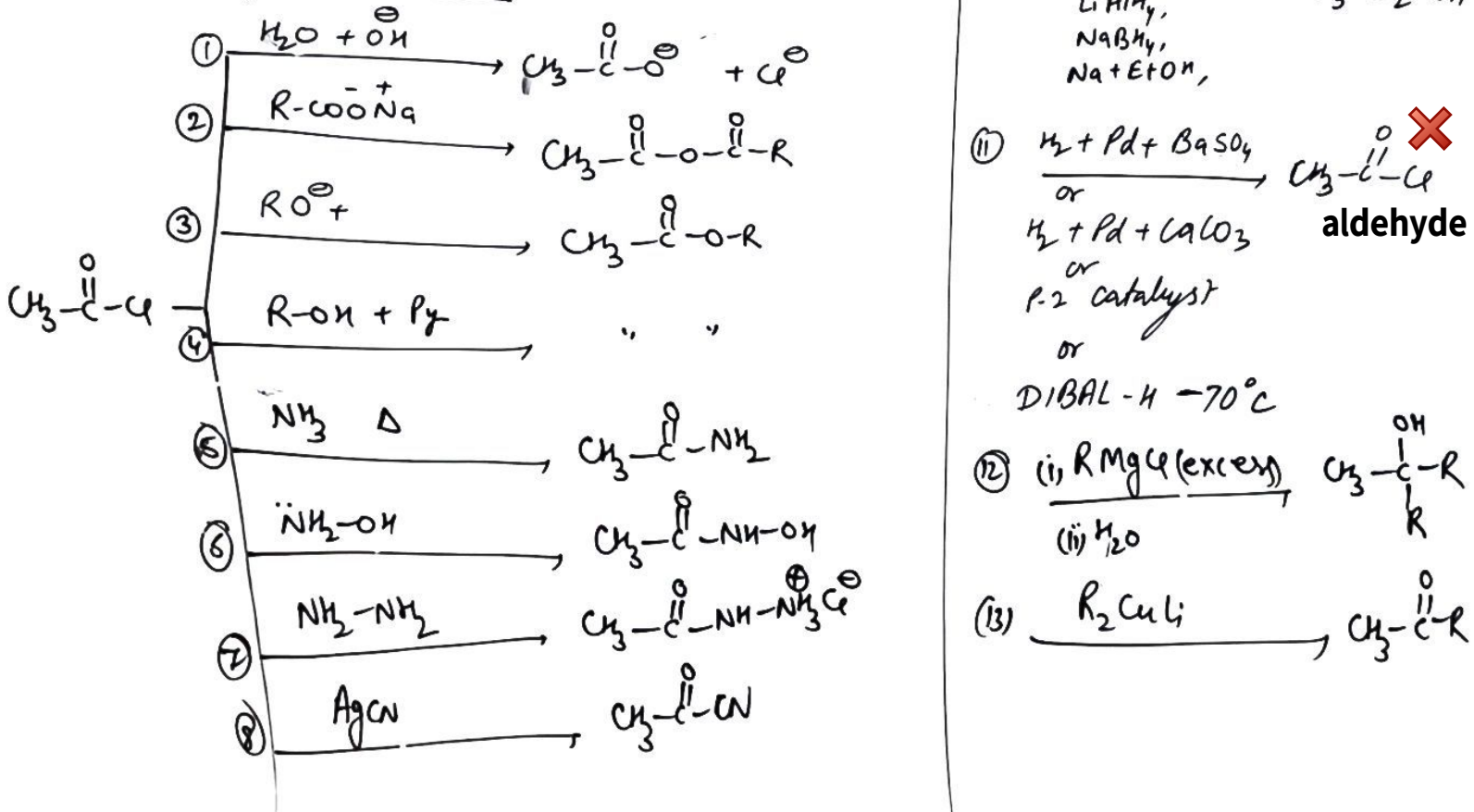
④ $R-CH=C=O \xrightarrow{HCl} R-CH_2-\overset{O}{\overset{\|}{C}}-Cl$

CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

(I) ACID HALIDE

Reaction of acid halides.

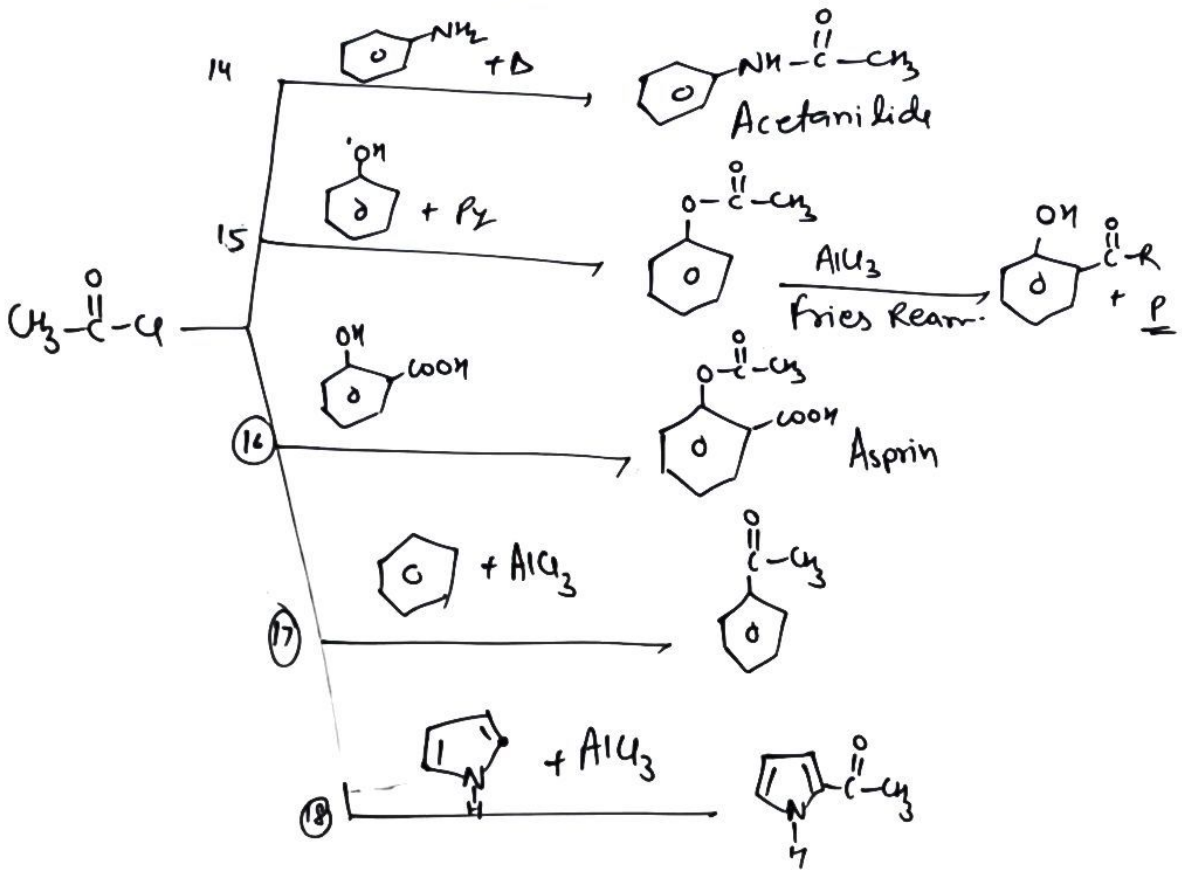


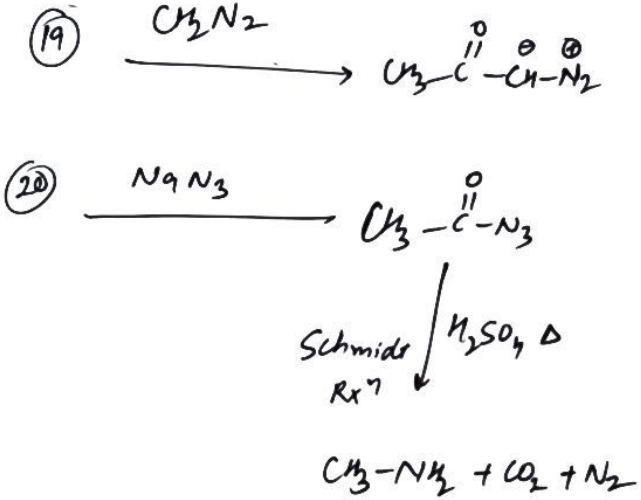
CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

(I) ACID HALIDE

Reaction of acid halides.





CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

(II) Carboxylic anhydride:-

Formation:-

① $R-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{P_2O_5} R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$

② $R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow{R'COO^{\ominus}} R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R'$

③ $R-CH=C=O \xrightarrow{R-COOH} R-CH_2-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$

★ ④ $R-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-R \xrightarrow{R''-COOOH} R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$

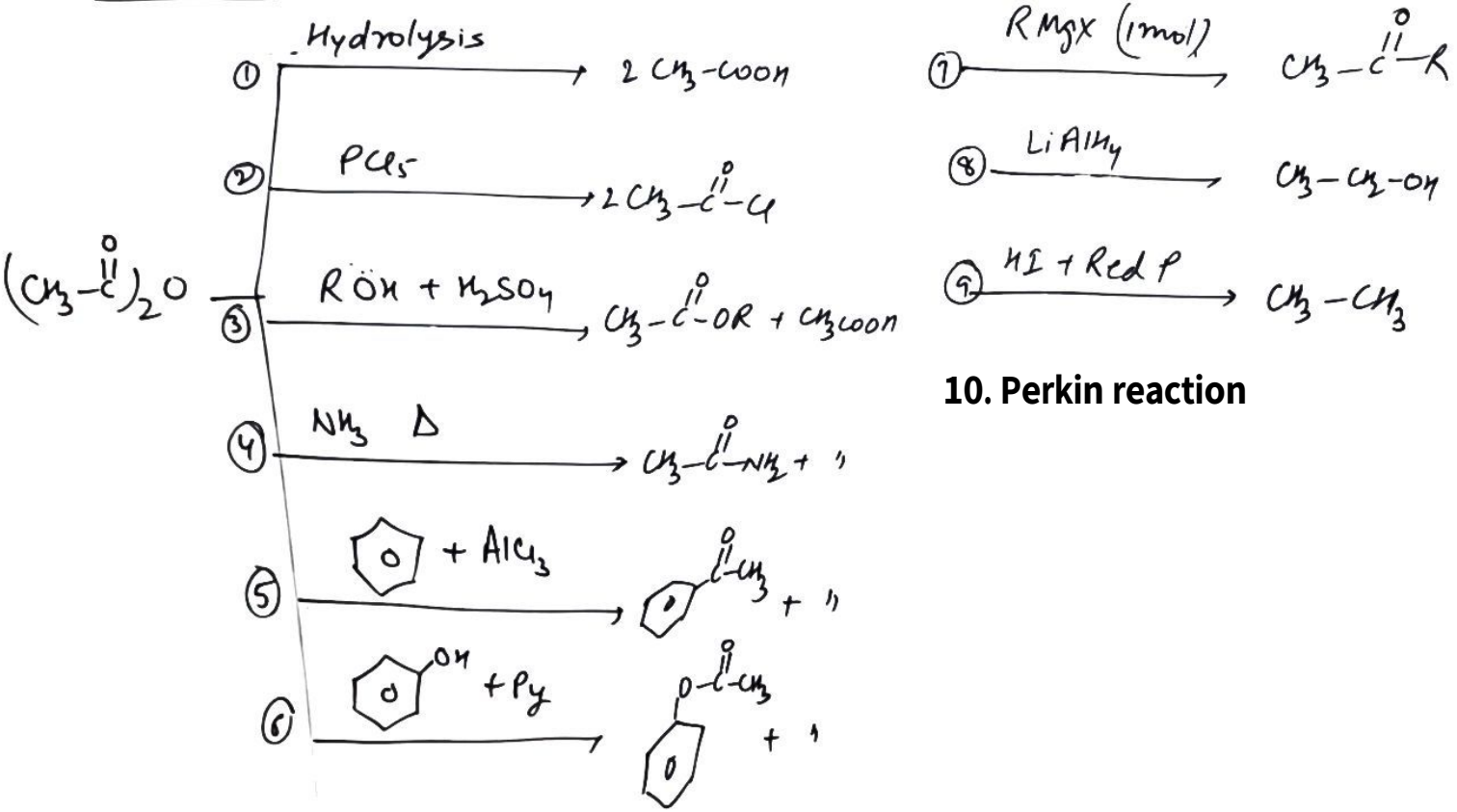
Baeyer-Villager oxidation

CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

(II) Carboxylic anhydride:-

Chemical Rx^n



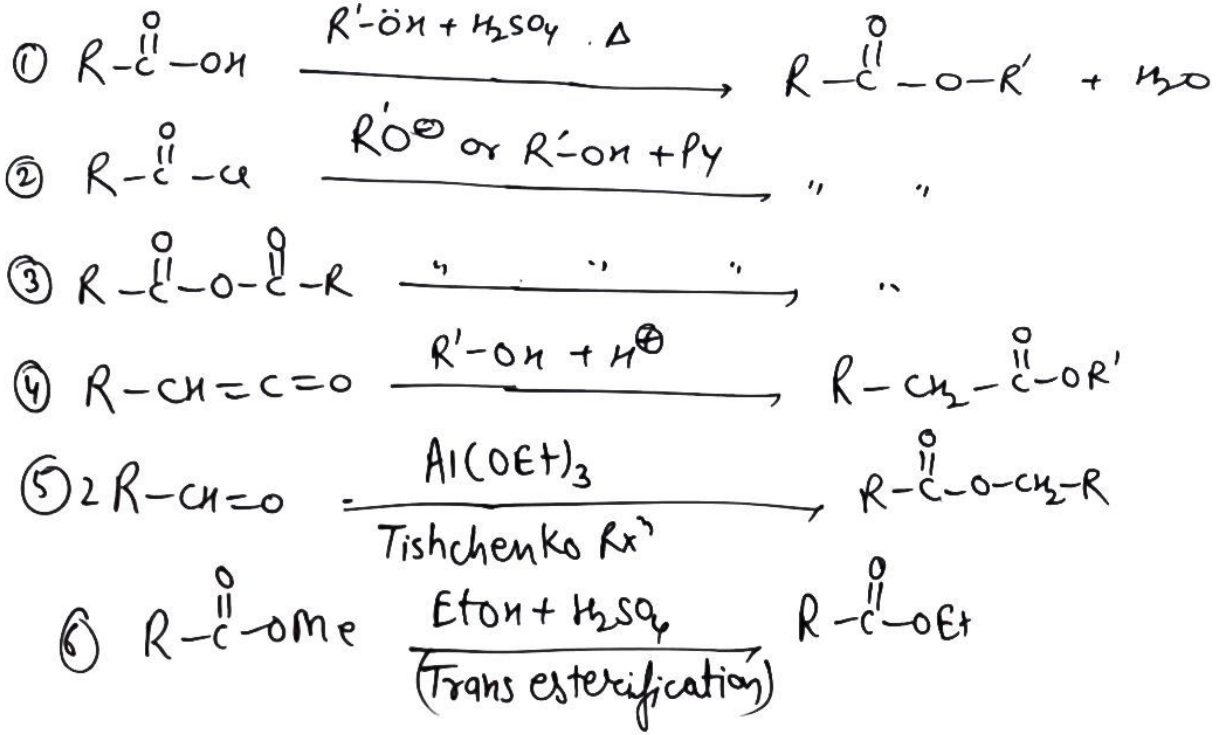
10. Perkin reaction

CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

(III) Ester

Formation



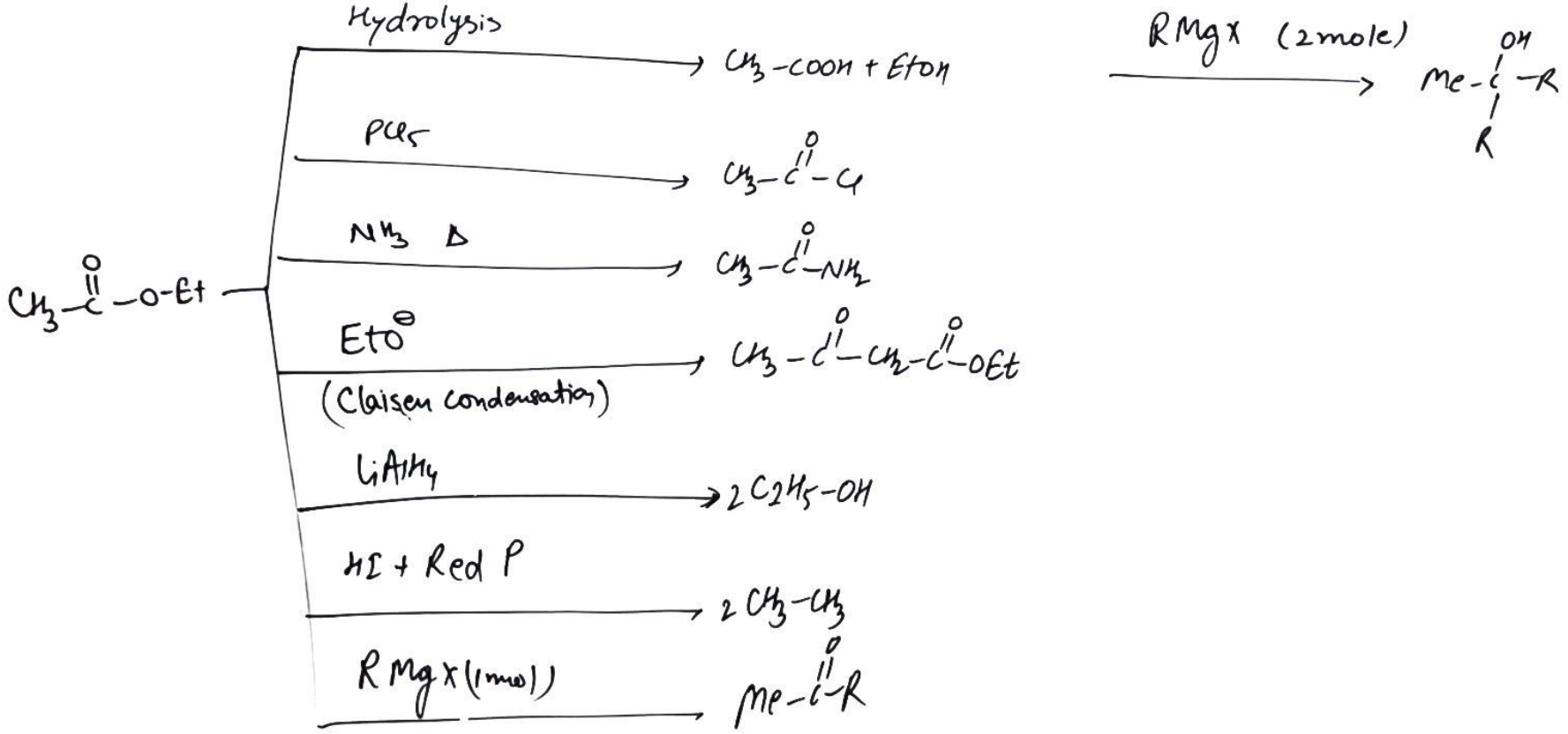
7. Baeyer Villager Oxidation

CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

(III) Ester

Chemical Rx^n

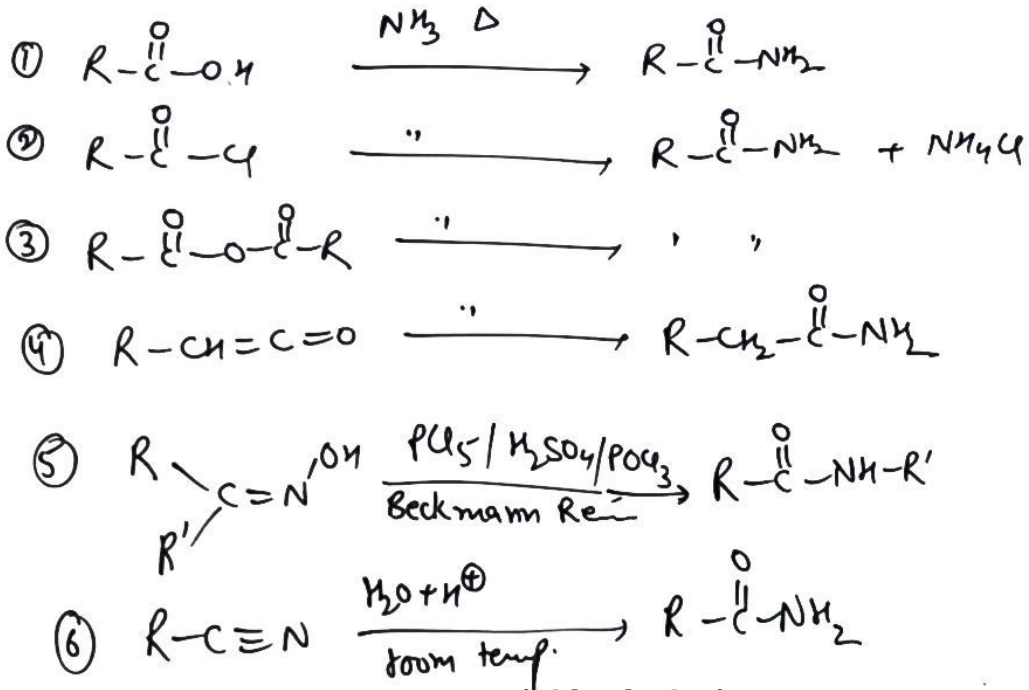


CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

IV Amide

formation



partial hydrolysis

CARBOXYLIC ACID AND DERIVATIVES OF CARBOXYLIC ACID

DERIVATIVE OF CARBOXYLIC ACIDS

IV Amide

Chemical Rxn

① $R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{H_2O + H^{\oplus}} R-\overset{O}{\overset{\|}{C}}-OH + \overset{\oplus}{N}H_4$

" $\xrightarrow{H_2O + \overset{\ominus}{O}H} R-\overset{O}{\overset{\|}{C}}-O^{\ominus} + NH_3\uparrow$

② $R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{LiAlH_4} R-CH_2-NH_2$

$\xrightarrow{HI + Red\,P}$ " "

③ $R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{MeMgX} R-\overset{O}{\overset{\|}{C}}-NH-MgX + CH_4$